

Day 9: Star Schema & Dimensional Modeling

📌 Definition to Remember

A **Star Schema** is the gold-standard dimensional data modeling architecture for Power BI where a central **Fact Table** (containing numeric measures and keys) is surrounded by multiple **Dimension Tables** (containing descriptive business context).

1. Fact Tables vs Dimension Tables

Feature	Fact Table	Dimension Table
Content	Quantitative metrics, transactions, numbers.	Descriptive text attributes, categories, hierarchies.
Examples	<code>Fact_Sales</code> , <code>Fact_Orders</code> , <code>Fact_Inventory</code> .	<code>Dim_Customer</code> , <code>Dim_Product</code> , <code>Dim_Date</code> , <code>Dim_Store</code> .
Granularity	Transactional event level (e.g., one row per line item).	Entity level (e.g., one row per unique customer).
Row Count	Millions / Billions of rows (Tall & Narrow).	Hundreds / Thousands of rows (Short & Wide).

2. Star Schema vs Snowflake Schema

```
STAR SCHEMA (Recommended for Power BI):
    [Dim_Product]    [Dim_Customer]
      \              /
      [Fact_Sales]
      /              \
    [Dim_Date]      [Dim_Store]
```

```
SNOWFLAKE SCHEMA (Normalized dimensions – sub-tables):
[Dim_Category] → [Dim_SubCat] → [Dim_Product] → [Fact_Sales]
```

3. The Dedicated Date Dimension Table (`Dim_Date`)

- Mandatory for all Power BI models to support accurate DAX Time Intelligence.
- Must cover full contiguous years without missing dates.
- Key columns: `Date` , `Year` , `MonthNumber` , `MonthName` , `Quarter` , `DayOfWeek` , `FiscalYear` .

★ Must-Master Skills & Interview Points

1. Why Power BI's VertiPaq engine is heavily optimized for Star Schema architectures.
2. Why Snowflake schemas should be flattened into a Star Schema in Power Query.
3. Why every professional BI model must have a dedicated, marked Date table.

⚡ Quick Recall

Central Fact Table + Surrounding Dimension Tables + Dedicated Date Dimension = Peak Power BI Performance